# OS UNIT4 ASSIGNMENT

https://github.com/Srujanakenche/PES1UG24CS227-pes-vcs

**PHASE 1:**

**Screenshot 1A** — capture ./test_objects output (shows all PASS)

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % make all
gcc -Wall -Wextra -O2 -I/opt/homebrew/opt/openssl@3/include -c tree.c -o tree.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
gcc -o test_objects test_objects.o object.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto -L/opt/homebrew/opt/openssl@3/lib
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ./pes init
Initialized empty PES repository in .pes/
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

**Screenshot 1B** — capture the sharded directory listing

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % find .pes/objects -type f

.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

**PHASE 2:**

**Screenshot 2A** — capture ./test_tree output (shows all PASS)

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ./test_tree

Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

**Screenshot 2B** — capture the xxd hex dump of the tree object

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % ulimit -s unlimited
rm -rf .pes
./pes init
echo "hello" > file1.txt
./pes add file1.txt
./pes commit -m "test"
find .pes/objects -type f | while read f; do if head -c 4 "$f" | grep -q "^tree"; then echo "$f"; xxd "$f" | head -20; break; fi; done


Initialized empty PES repository in .pes/
Committed: d6b371f8ffe1... test
.pes/objects/09/b576c2eb51b78fcf9b8493a0909522a75f8ee86df0e926312deeb5fbe7dfb1
00000000: 7472 6565 2034 3900 3130 3036 3434 2066  tree 49.100644 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt.,..=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!........7
00000030: 6390 25de 576d b9eb b4                   c.%.Wm...
```

## PHASE 3:

**Screenshot 3A** — capture the full sequence output with status

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % rm -rf .pes hello.txt file1.txt file2.txt bye.txt
  ./pes init
  echo "hello" > file1.txt
  echo "world" > file2.txt
  ./pes add file1.txt file2.txt
  ./pes status

  Initialized empty PES repository in .pes/
  Staged changes:
    staged:     file1.txt
    staged:     file2.txt

  Unstaged changes:
    (nothing to show)

  Untracked files:
    untracked:  index.h
    untracked:  report_assign4.pdf
    untracked:  object.c
    untracked:  pes.c
    untracked:  .DS_Store
    untracked:  Makefile
    untracked:  test_objects
    untracked:  tree.h
    untracked:  test_objects.c
    untracked:  commit.c
    untracked:  test_tree
    untracked:  README.md
    untracked:  index.c
    untracked:  test_sequence.sh
    untracked:  .gitignore
    untracked:  test_tree.c
    untracked:  compile_flags.txt
    untracked:  pes.h
    untracked:  commit.h
    untracked:  tree.c
```

**Screenshot 3B** — capture the index file contents

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % cat .pes/index

  100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776766066 6 file1.txt
  100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776766066 6 file2.txt
```

## PHASE 4:

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % echo "Hello" > hello.txt
  ./pes add hello.txt
  ./pes commit -m "Initial commit"
  echo "World" >> hello.txt
  ./pes add hello.txt
  ./pes commit -m "Add world"
  echo "Goodbye" > bye.txt
  ./pes add bye.txt
  ./pes commit -m "Add farewell"
  ./pes log

  Committed: c65fc1be0410... Initial commit
  Committed: 7cee187aa8b0... Add world
  Committed: cd99e8c6fd2d... Add farewell
  commit cd99e8c6fd2d50d8e2c040a88c99dd80060fa9a4c37ab0c13f7968265b4350aa
  Author: PES User <pes@localhost>
  Date:   1776766205

      Add farewell

  commit 7cee187aa8b00e5f2599b971f696b1b450f811cb4c677046b1573e02a470e34b
  Author: PES User <pes@localhost>
  Date:   1776766205

      Add world

  commit c65fc1be0410b41a0827239123462cf9e00d41737579294a6871202c004f4bf1
  Author: PES User <pes@localhost>
  Date:   1776766205

      Initial commit
```

**Screenshot 4A** — capture ./pes log showing all 3 commits
**Screenshot 4B** — capture the full object store listing

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % find .pes -type f | sort

  .pes/HEAD
  .pes/index
  .pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
  .pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
  .pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
  .pes/objects/7c/ee187aa8b00e5f2599b971f696b1b450f811cb4c677046b1573e02a470e34b
  .pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
  .pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
  .pes/objects/c6/5fc1be0410b41a0827239123462cf9e00d41737579294a6871202c004f4bf1
  .pes/objects/cd/99e8c6fd2d50d8e2c040a88c99dd80060fa9a4c37ab0c13f7968265b4350aa
  .pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
  .pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
  .pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
  .pes/refs/heads/main
```

**Screenshot 4C** — capture both outputs

```
● bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % cat .pes/refs/heads/main
  cat .pes/HEAD

  cd99e8c6fd2d50d8e2c040a88c99dd80060fa9a4c37ab0c13f7968265b4350aa
  ref: refs/heads/main
```

## Final Screenshot:

```
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs % make test-integration

=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 606fa0560aed... Initial commit

Log after first commit:
commit 606fa0560aed930ceb586066c8821055f04e99fee9d1cc3b7dd8b5a7dc30511f
Author: PES User <pes@localhost>
Date:   1776766406

    Initial commit


--- Second Commit ---
Committed: 783353d17045... Update file.txt

--- Third Commit ---
Committed: 09d4e2a53c5e... Add farewell

--- Full History ---
commit 09d4e2a53c5e0cf85f0272d333934b3e61bfdb4cd4c66ab498cb0aecbeb11f13
Author: PES User <pes@localhost>
Date:   1776766406

    Add farewell

commit 783353d1704511baf9af82fd9d6d766425c71a92dd80c9360a44329981da7014
Author: PES User <pes@localhost>
Date:   1776766406

    Update file.txt

commit 606fa0560aed930ceb586066c8821055f04e99fee9d1cc3b7dd8b5a7dc30511f
Author: PES User <pes@localhost>
Date:   1776766406

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
09d4e2a53c5e0cf85f0272d333934b3e61bfdb4cd4c66ab498cb0aecbeb11f13

--- Object Store ---
Objects created:
      10
.pes/objects/09/d4e2a53c5e0cf85f0272d333934b3e61bfdb4cd4c66ab498cb0aecbeb11f13
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/60/6fa0560aed930ceb586066c8821055f04e99fee9d1cc3b7dd8b5a7dc30511f
.pes/objects/78/3353d1704511baf9af82fd9d6d766425c71a92dd80c9360a44329981da7014
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
bvell@BhaveshVellurus-MacBook-Air PES1UG24CS227-pes-vcs %
```

**ANALYSIS QUESTIONS**

## 5. Analysis of Object Storage

- **5.1 Why is the object store sharded into subdirectories (e.g., `.pes/objects/d5/...`)?** The object store uses sharding to maintain file system performance. Storing thousands of files in a single directory can cause significant slowdowns during file lookups and writes on many file systems. By using the first two hex characters of the hash (e.g., `d5`, `2a`, `25`) as a subdirectory, the system distributes files across 256 possible directories, ensuring each directory remains manageable.
- **5.2 How does this sharding strategy affect the path where an object is stored?** The path is constructed by taking the first two characters of the SHA-256 hash as the parent directory and using the remaining 62 characters as the filename. For example, in your output, the object with hash `d58213f5...` is stored specifically at `.pes/objects/d5/8213f5....`

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## 6. Analysis of the Staging Area

- **6.1 What is the role of the `index` file in the staging process?** The `index` file acts as a mandatory "middle ground" or staging area between the working directory and the final commit. It tracks which files have been prepared for the next snapshot without requiring the system to scan the entire project directory every time a commit is made.
- **6.2 What information is stored for each entry in the `.pes/index` file?** Based on your `cat .pes/index` output, each entry contains:
  - **File Mode**: The permissions (e.g., `100644`).
  - **Hash**: The unique 64-character SHA-256 blob identifier for the file content.
  - **Timestamp**: The `mtime` or last modification time (e.g., `1776766066`).
  - **File Size**: The size of the file in bytes.
  - **Path**: The relative path/name of the file (e.g., `file1.txt`).